

Atharv Vatsal

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EDUCATION

Vellore Institute of Technology

Vellore, Tamil Nadu

Bachelors of Technology – Computer Science and Engineering

Expected 2027

Specialization in Artificial Intelligence and Machine Learning

- Specialized coursework in Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, System Design, and Computational Mathematics.
- Hands-on experience in digital system design, embedded C/C++, and real-time computational problem-solving.**
- Passionate about machine learning, deep learning, and computer vision**, with projects spanning text-to-voice conversion, predictive modelling, and AI ethics.
- Strong foundation in data structures, algorithms, and computational mathematics**, focusing on efficient AI model optimization and high-performance computing.
- Developed multiple ML-driven projects, including **YOLO-based can detection, text-to-speech systems, predictive modeling, and statistical analysis.**

Dhauladhar Public School

Dharamshala, Himachal Pradesh

10+2 Certification HPBOSE

May 2023

State Merit #51 | 93.3%

Sacred Heart Sr. Sec. School

Dharamshala, Himachal Pradesh

10th Certificate CICSE

July 2021

First Division | 90.7%

EXPERIENCE

Intern – Department of Digital Technologies & Governance, Government of Himachal Pradesh

May, 2025 – July, 2025

- Worked on **machine learning and data science initiatives** supporting digital governance and public data management.
- Designed and implemented **AI/ML models** to optimize government workflows, enhancing decision-making efficiency across departments.
- Conducted **data cleaning, preprocessing, and exploratory data analysis (EDA)** on large-scale administrative datasets to identify key operational insights.
- Built and evaluated **predictive models** (using Python, Scikit-learn, and Pandas) for trend forecasting and resource allocation.
- Developed **interactive dashboards** and visualization tools using Power BI and Plotly to communicate findings to non-technical stakeholders.
- Applied **computer vision techniques** for automating document processing and image-based data validation.
- Collaborated with cross-functional teams to **translate governance challenges into data-driven solutions** aligned with policy goals.
- Delivered **data-driven reports and presentations** to senior officials, contributing to digital transformation strategies and improved citizen service delivery.

PROJECTS

RiskGrid - AI-Powered Spatio-Temporal Crime Prediction System

Developed an end-to-end machine learning system that predicts crime patterns 24 hours in advance by transforming 7.8+ million historical crime records into actionable intelligence for law enforcement resource allocation. The system divides Chicago into a 100m × 100m spatial grid and employs an ensemble of XGBoost, Random Forest, and LightGBM models to forecast high-risk zones with 87.4% accuracy. To address severe class imbalance (92% zero-crime cells), I implemented SMOTE oversampling combined with stratified cross-validation and custom cost-sensitive loss functions. Feature engineering involved creating 60+ predictive variables including temporal patterns (hour/day/season), spatial autocorrelation (k-nearest neighbour crime density), rolling crime averages, and lag features. For scalability, I leveraged DASK for distributed processing of geospatial joins and vectorized Pandas operations for efficient aggregation. The system features a dual architecture: a Flask-powered real-time prediction endpoint and an offline batch processing pipeline for historical analysis. The online dashboard with Folium-based heat maps, Plotly time-series visualizations, and feature importance analysis for interpretability. Model training utilized GridSearchCV for hyperparameter tuning, MLflow for experiment tracking, and ensemble stacking to maximize predictive performance. The modular pipeline supports plug-and-play integration of LSTM networks for temporal forecasting and easy adaptation to other cities by reconfiguring grid boundaries. The system demonstrates potential to prevent 1,500+ crimes annually, transforming reactive policing into proactive crime prevention.

- Technologies Used:** Python, Scikit-learn, XGBoost, LightGBM, Pandas, Dask, GeoPandas, NumPy, Streamlit, Folium, Plotly, FastAPI, SMOTE, MLflow, GridSearchCV, Feature Engineering, Ensemble Learning, Geospatial Analysis, Time-Series Forecasting

HP Police ProReportStream AI

Collaborated with the Himachal Pradesh Police to design and implement an end-to-end automation pipeline that transforms unstructured WhatsApp battalion reports into standardized Excel dashboards. The system combines a hybrid regex and DistilBERT NLP model to enable robust field extraction, even with noisy inputs, inconsistent formatting, and multilingual text. To ensure data integrity, I engineered validation and anomaly-detection layers, alongside a custom Excel formatting engine built with XlsxWriter for pixel-perfect alignment, cell styling, and dynamic table generation. Performance was optimized using vectorized Pandas operations, asynchronous I/O for bulk report handling, and caching of intermediate NLP outputs. By integrating transformer-based models with rule-based fallbacks, the pipeline achieved over 98% extraction accuracy. Its modular, extensible architecture supports future plug-and-play upgrades such as multilingual BERT and domain-specific fine-tuning. Ultimately, the system reduced report preparation time from 2–3 hours of manual entry to under 10 seconds, significantly accelerating decision-making in law enforcement.

- Technologies Used:** Python, Pandas, Streamlit, Regex, Transformers (Hugging Face – DistilBERT), NLTK, XlsxWriter, NumPy, Async I/O, Docker (planned), Cron/Airflow (planned), Automation

TheCanspacy | Can Detection and Classification

Developed a computer vision application to detect and classify branded cans (Red Bull, Monster, Pepsi, Coca-Cola, etc.) using YOLO object detection. The project focuses on real-time detection, model optimization, and continued enhancements. Enhances automated product recognition for inventory management and collection tracking. Optimized detection performance by fine-tuning YOLO parameters, modifying YOLO source code and augmenting training data. Currently enhancing the application with features like custom script execution and easy model selection and cross platform integration along with a user-friendly GUI.

- Technologies Used:** Python, OpenCV, TensorFlow, YOLOv8, Data Augmentation, Model Optimization.

VOLUNTEER EXPERIENCE & EXTRACURRICULARS

Design and Media – Riviera'25

ProShow Media Photographer And Fest Photography Lead

January, 2025 – March, 2025

- Served as the **official concert photographer** for the ProShow, capturing **high-profile performances and celebrity appearances**, including Shreya Ghoshal, Jonita Gandhi, Neeti Mohan, and Sonu Sood. Managed **live event photography**, ensuring high-quality coverage under dynamic lighting conditions. Contributed to media production, social media promotions, and post-event content creation.

VitrendZ

Director of Photography

March, 2025 - Present

- Leading the visual storytelling for VITrendz, VIT's largest digital media platform, by capturing the essence, vibe, and culture of the college through impactful photography and videography. Responsible for curating engaging content, covering major campus events, and shaping the visual identity of the platform.

The Photography Club

Photographer

January, 2024 – February, 2025

- Captured **premium events** across campus, including SAE's **Stunt Show**, **Design Odyssey**, **GDSC's Hexathon**, and other **high-profile technical and cultural events**. Specialized in **action photography**, **dynamic event coverage**, and **post-processing** to deliver professional-quality visuals for official media and promotions.

SKILLS

- **Programming Languages:** Java, Python, C, C++, Kotlin, BASH Scripting, SQL, HTML, CSS, JavaScript, PHP(Basics), MATLAB, R, Linux Scripting
- **Artificial Intelligence & Machine Learning:** Numpy, Pandas, Matplotlib, Seaborn, TensorFlow, Scikit-learn, Computer Vision
- **Data Structures & Algorithms:** Competitive programming, Optimized algorithm design
- **Cloud & DevOps:** AWS (Basics), Firebase, Streamlit
- **Version Control & Collaboration:** Git, GitHub
- **Mathematical & Computational Skills:** Linear Algebra, Probability and Statistics, Calculus, Discrete Mathematics, Differential Equations, Complex Mathematics
- **Software Architecture & Design:** OOP, System Design, API Integration
- **Cybersecurity & Authentication:** Secure Data Handling, Encryption Basics

Problem-Solving & Analytical Thinking | Leadership & Teamwork | Communication & Public Speaking | Time Management & Multitasking | Creativity & Innovation | Attention to Detail